

13F Positioning Factors — Memo

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August 2026

Compressed from the full technical report (`docs/paper/main.pdf`, 30 pp.), which carries every construction, pre-registration, surface, negative result, and one explicit retraction. Reproduction: `python bootstrap.py` & `python run_experiments.py` from a clean clone (EDGAR needs only a contact e-mail; prices auto-fetch).

1. Factor hypothesis

13F filings are stale (up to 45 days), long-only, quarterly: most of their content is already in prices. Our premise: a premium can only live where the filing reveals an action that was *costly to take* or *impossible to avoid*; ownership *states* — crowding, concentration, breadth, days-of-ADV, factor tilt — carry no direction. The premise was stress-tested against a six-class taxonomy (17 catalogue signals with literature anchors, 5 pre-registered interactions, 9 smart-money designs, 15 crowding formulations, latent-structure and event-clock studies; ~60 experiments, every one with its pre-registration in the script docstring) and survived as two factors and two empirical laws. **Law 1: states have no arrow** (15 crowding formulations dead at every clock — graph diffusion, centrality, absorption, strategy-crowding included; states earned only two jobs: denominators and risk conditioning). **Law 2: distress is information when observed, never predicted** (contemporaneous forced selling $t \approx +10$; every anticipation design — daily synthetic stress, quarterly Coval–Stafford-style flow forecasts (OOS rank-corr -0.015), crash cascades — failed its mechanism gate or died on the full panel).

NCB — New-Conviction Breadth (long). Per stock, the share of filers opening or growing ($\geq 1.25\times$, split-adjusted by a holder-consensus modal-ratio detector, validated 6/6) a position in the *top decile of their own book*; residualized on $\log\text{mktcap} + \log\text{ADV}$ because a formal placebo (dart-throwing value-weighted managers) shows the raw count is mechanically size-loaded. Mechanism: top-of-book room is made by displacing existing conviction — expensive, deliberate, and impossible to produce by indexing or inflows; counting it across independent managers aggregates only paid-for opinion. Results: $+1.9\%/q$ spread, $t = +3.44$, Sharpe ≈ 1.0 , IC $+0.023$ (fundamental law: $0.023 \times \sqrt{1900} \approx 1.0$ — breadth, not per-name accuracy). Only entry clearing the catalogue-wide Bonferroni bar ($t \geq 3.2$, 34 tests); 8 implementation replications in a $t = 3.3\text{--}4.0$ band across two data vintages; survives random 50% coverage halvings ($t = +3.45/+2.79$); joint Fama–MacBeth premium $t = +3.04$ holding six classical characteristics and three other 13F signals simultaneously; unabsorbed by IPCA latent factors (spanning $t \approx +6$). Hyperparameters (0.90 weight-quantile / 1.25 growth / 5 min holders) fixed a priori, never tuned; the full $4\times 3\times 3$ surface, mapped later under a coupled block-bootstrap 5th-percentile protocol (arXiv:2510.12725), is *uniformly positive* out-of-sample (all 36 configs OOS Sharpe $+1.05$ to $+2.53$; in-sample ranks OOS at $+0.53$, so selection is meaningful; the a-priori default sits mid-surface, conservative).

FSS — Forced-Sale Supply (short). Implied flow $= \text{AUM}_t / \text{AUM}_{t-1} - (1 + r^{\text{book}})$ separates client/leverage money from market growth. Validated on behavior, not returns: flow $< -10\%$ managers sell **34.5%** of remaining shares the *following* quarter vs 23.1% for healthy ($t \approx +10$ at every cut from -5% to -20% — monotone curve, not a fitted number; selling is $\approx$ pro-rata across liquidity, correlation -0.02). Stock supply $= \sum_{\text{dis}} |\text{flow}| \cdot \text{AUM} \cdot w / \text{ADV}$. Price: index-hedged, all 15 threshold $\times$ basket configs are flat 2013–20 (Sharpe -0.1 to -0.5) and pay 2021–25 ($+0.6$ to $+1.0$, t to $+4.0$) — a *regime* (degrossing/rates), not a hyperparameter: IS-to-OOS rank correlation across configs is -0.81 , so in-sample selection is refused and the a-priori spec ships. Benchmark honesty: an early “ $-4.5\%/q$ vs universe mean” figure was skewness (the cross-sectional mean sits $\sim 4\%/q$ above the median); the hedged number is the true one. The pressure autopsy explains why generic flow-pressure fails (impact realized inside the filing quarter; orthogonalized to momentum/reversal nothing remains) and why FSS works: liquidation *continues* past disclosure only when selling is forced — condition on the agent’s constraint, not the asset’s state. The pre-registered pressure $\times$ vol interaction (E4) died properly: smoke $\rightarrow t = +2.06$ old universe $\rightarrow +0.77$ on complete books; its identification autopsy (vol vs ADV vs mktcap) had already returned “no formal claim.”

2. Scope decisions and deliberate exclusions

2013Q2–2025Q4: the window where two independent ingestion paths (DERA structured datasets; raw EDGAR parsing) overlap and cross-validate (row counts and value totals per sampled filing; $<95\%$ agreement halts the pipeline). **Universe: the full filer census** (2,793 $\rightarrow$ 7,693) — a measured choice, not a default: activity/concentration/patience/performance filters hurt (up to $t = -3.25$) or do nothing; the passive question closed on three fronts (index complexes = $1/3$ of AUM, 1% of conviction votes, removal moves the spread 6 bps; ETF

purge inside books changes vote-sets of 25% of managers yet leaves the ranking at correlation 0.974 — uniform contamination is innocuous, unlike splits, which are differential and required the detector). **Assets:** common equities, mapped current ticker, \$1 floor; options excluded (no deltas), ETF instruments excluded (never in the price panel — 1 of 7,886 audited ETF ids maps). **Not pursued, with evidence:** smart-money weighting (0-of-9; skill persistence 0.25 and $\perp$ holdings — reweighting the electorate is reweighting at random); imitation of books (the median \$1B+ filer loses $-3.3\%/yr$ to the S&P before fees, 13% positive-IR: Bessembinder composition arithmetic — the alpha is in flows *between* managers, not in the books); window dressing (both pre-registered signs wrong; placebo follows thesis; a Q4-only reversal of -1.9% is an 11-observation footnote); amendment-following (materials price *against* the correction, $t = -3.5$); daily stress anticipation (**retracted**, below).

3. Data engineering and point-in-time discipline

Bitemporal store: every row carries period and EDGAR acceptance timestamp; research reads holdings only through `as_of(period, knowledge_date)`, materializing current and prior quarter at the same vantage — late filings, amendments, and restatements cannot leak. Amendment semantics from cover pages (restatement replaces; NEW HOLDINGS — lapsed confidential treatment — adds; missing = conservative restatement; an amendment is a version, never a competing filing). Value scale detected per filing (the 2022–23 thousands/dollars cutover keys on filing date). Identity: union-find on explicit evidence; name similarity only proposes; merges need a human entry. **The D+45 clock is measured, not assumed:** across 254,630 filings, median delay 41 days, 49.9% by D+40, 89.4% by D+45, 98.1% by D+60, p99 = 128d; lateness is sticky (41% repeat); per-name ragged entry adds $+0.2\%/yr$ ($t \approx 0$) over the batch clock. Forward returns start the first trading day *after* decision. **The crosswalk is a result:** CUSIP→issuer-name→current ticker; renames correct by construction, recycling fails closed (4.5% flagged), CUSIP reassignment measured at 0.22% of exit flow. A value-weighted audit found **46.4% of book value unmapped** (EDGAR abbreviations defeat exact matching precisely on mega-caps; count-weighted coverage had looked fine — audits must be value-weighted). Fix: top-250 CUSIPs hand-verified (63→85.4% of non-ETF value), then a **24-run re-verification campaign:** 21 verdicts stable; two instruments *strengthened* (flow contrast $t = 13.8 \rightarrow 20.9$; synthetic short interest vs real, corr $0.47 \rightarrow 0.55$); one result **retracted** — a daily stress-anticipation short ($t = -2.16 \rightarrow +0.37$ on complete books; an eligibility-selection artifact the funnel had already declined to promote). **Survivorship measured:** the price panel conditions on survival to download (nothing dies inside a window — verified); the unmapped tail is 33% of eligible-instrument value with an NCB vote rate only $+0.002$ different ($t = +2.4$); the delisting truncations (bankruptcies missing from the no-vote quintile, takeover premia from conviction names) both cut *against* the measured spread. CRSP closes both ($85.4 \rightarrow \sim 98\%$).

Table 1: The re-verification campaign in brief (full table in the report): pre-registered expectation was levels move where mega-caps matter, verdicts stand.

result	old universe	expanded	verdict
NCB (residualized)	$t = +3.67 / +3.83 / +3.27$	$t = +3.44$	in band; 8th replication
flow instrument	$t = +13.8$	$t = +20.9$	stronger with complete books
synthetic short interest	corr $+0.47$	$+0.55$	stronger
FM joint λ (NCB / dbreadth)	$+3.26 / +4.62$	$+3.04 / +4.01$	holds
pressure $\times$ vol flicker	$t = +2.06$	$+0.77$	died on fuller data
median fund excess	$-5.0\%/yr$	$-3.3\%/yr$	level moved as predicted
daily stress-anticipation	$t = -2.16$	$t = +0.37$	retracted (artifact)

4. Evaluation protocol and backtest methodology

One protocol, fixed ex-ante: quintile portfolios, Q5–Q1 on forward quarterly returns (sums of daily simple returns — quintile-neutral convention), Newey–West t (4 lags on non-overlapping series = conservative), Spearman IC, seeded random tie-breaks (alphabetical ties fabricate signal at mass points). **Why the EW base:** the evaluation question is “does the ranking order returns?”, under which every name is one observation — value-weighting collapses effective N onto a handful of mega-caps and breaks the $IC\sqrt{N}$ arithmetic the signals live on; EW is the *measurement* convention, cap-weighting the *implementation* convention (engine and LO below), VW reported as the bridge (indicative: its weight panel’s corrupt tail is documented). **Why residualize:** a formal placebo — dart-throwing value-weighted managers — decides neutralization per signal, ex ante: if uninformed books produce the size/liquidity correlation mechanically, the headline is the cross-sectional residual on $\log mktcap + \log ADV$; otherwise raw. One rule, two opposite unmaskings: NCB raw is *nothing* ($t = -0.9$; the mechanical large-cap drag masks the information — the residual is where the signal lives, $t = +3.5$), while days-of-ADV raw replicates its published point and dies residualized ($t = -0.4$) — the published crowding effect *was* size/liquidity. Verdicts are paired quarter-by-quarter deltas on common samples; families carry Bonferroni bars with iteration counts; post-hoc entrants are candidates, never promotions; smoke never decides (four smoke mirages died on full panels). **Engine:** tradable universe (\$4/\$5M

ADV/252d), 20% tails, 40-name leg minimum, rebalance at each signal's decision dates, **5 bps/side** (in line with realized institutional costs; quarterly trading, measured one-way turnover $3\text{--}7\times/\text{yr}$, and the tradability filter is the liquidity control keeping slippage second-order; impact belongs to capacity, not backtests), **50 bps borrow** (at 200 bps hard-to-borrow the FSS leg loses $\approx 1.5\%/yr$, Sharpe $0.85\rightarrow 0.73$ — survives). Combo legs weighted *equally by design* (no-optimization default).

5. Results

Table 2: Net production results (5 bps/side; 50 bps borrow; expanded universe). Cost = measured commission drag (net minus zero-fee run); turn = implied one-way annual turnover. LO rows: excess/TE/IR vs own benchmark; the cap-weighted 500 proxy tracks the actual S&P at 2.7% TE ($-0.6\%/yr$ drift).

strategy	ann. net	vol	Sharpe	maxDD	cost (bps/y)	turn
NCB+FSS book	+6.2%	6.0%	+1.02	−10.6%	67	6.7×
FSS short leg	+10.4%	12.2%	+0.85	−26.1%	65	6.5×
NCB (residualized)	+2.9%	6.7%	+0.42	−18.7%	65	6.5×
FM score (10 predictors)	+4.0%	16.9%	+0.24	−62.1%	43	4.3×

Long-only expression. A naive EW long basket against the S&P is condemned by the composition arithmetic above (the Bessembinder tax the median 13F book itself pays); the institutional design tilts a genuine cap-weighted model — top-500 by repaired market cap, validated at 2.7% TE to the actual index — by the combo rank, MSCI-style multiplicative exponential (positivity preserved, unscored names stay at benchmark weight). IR $\approx 0.5\text{--}0.6$, roughly *flat in tilt strength*: the signature of a scalable signal, with k shipped as a tracking-error budget (Table 3, Figure 1).

Table 3: Long-only cap-weighted model and combo tilts, net of 5 bps/side, quarterly. Excess/TE/IR and relative maxDD (worst peak-to-trough of the log wealth ratio) vs the tilt's own benchmark; last row vs the actual S&P.

portfolio	total net	vol	Sharpe	maxDD	excess	TE	IR	rel. DD
S&P 500 (actual index)	+14.8%	17.1%	0.86	−41%	—	—	—	—
cap-weighted 500 (proxy)	+14.1%	18.0%	0.78	−45%	—	—	—	—
combo tilt $k=0.5$	+14.6%	18.2%	0.80	−45%	+0.4%	0.9%	0.49	−2.5%
combo tilt $k=1$ (default)	+15.1%	18.4%	0.82	−45%	+1.0%	1.8%	0.53	−5.7%
combo tilt $k=2$	+16.4%	19.2%	0.85	−44%	+2.2%	4.0%	0.57	−13.7%
combo tilt $k=3$	+17.7%	20.4%	0.87	−44%	+3.6%	6.2%	0.58	−21.2%
tilt $k=3$ vs S&P (actual)					+2.9%	7.3%	0.40	−25.5%

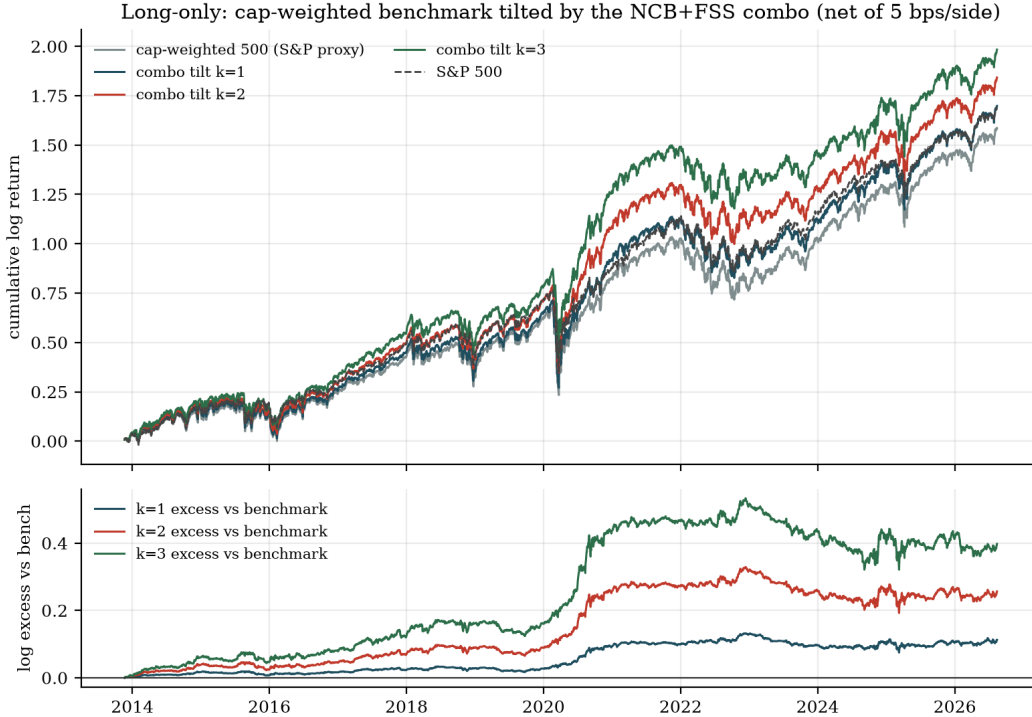


Figure 1: Long-only benchmark tilt, net (log cumulative). Top: levels; bottom: log excess vs the benchmark by tilt strength.

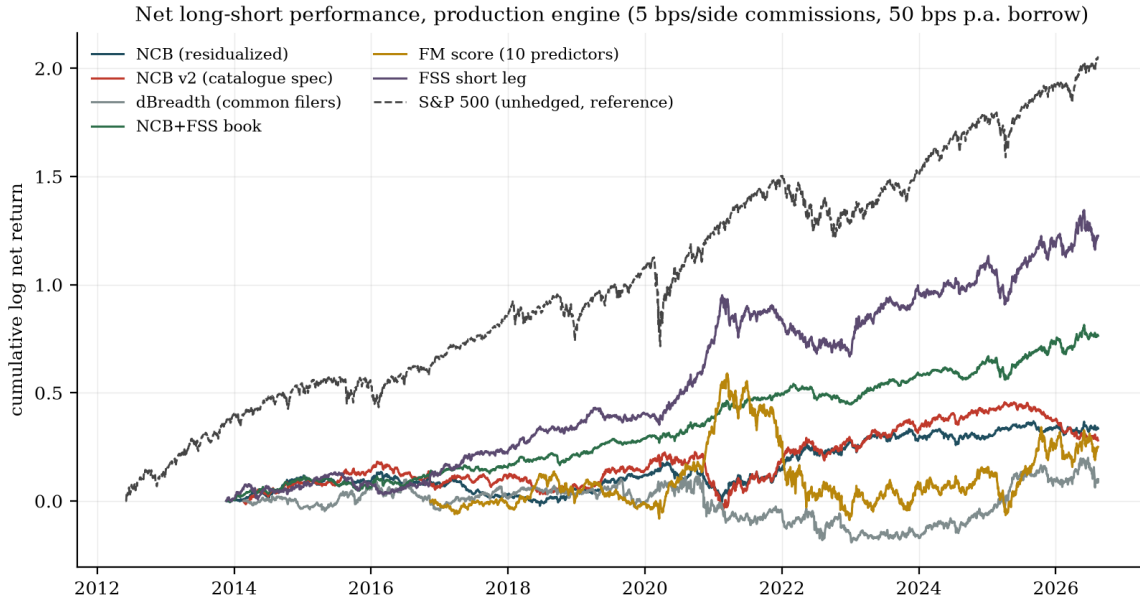


Figure 2: Net long-short performance, production engine (log cumulative; 5 bps/side, 50 bps p.a. borrow, expanded universe).

Combining models. Three combiner schools raced strictly OOS on ten pre-registered predictors: Fama–MacBeth/Lewellen wins the research protocol (Sharpe 0.86 vs IC-weighted 0.69 and naive 0.74; A–C paired $t = +1.96$); joint marginality: NCB $\lambda t = +3.04$, breadth-change $t = +4.01$ (an *ingredient*: weak solo, strong jointly — it also replicates Chen–Hong–Stein at $+1.65\%/q$). The engine decomposition is a result: the FM book nets 0.24 with a -62% drawdown because its two biggest premia (size $t = +4.45$, liquidity $t = +4.39$) live outside the tradable set — on the tradable universe, residualized, the FM ranking retains *nothing* ($t = -0.70$); the tradable alpha is its 13F components, hence the NCB+FSS book. IPCA (Kelly–Pruitt–Su, verified against the authors’ package): W_α cannot reject at $T=50$ ($p = 0.4-0.6$, power); unrestricted OOS Sharpe 0.97 beats FM paired ($t = +2.08$) but is a post-hoc entrant $\Rightarrow$ candidate, FM remains production; the spanning answer stands (no latent factor absorbs NCB). An ML cross-check (ridge/kNN/boosting/RF on the 12 change variables, walk-forward, market+size-residualized rank target) finds the family predicts ($t = +2.6$ to $+3.4$) but NCB alone *beats the 12-variable model* ($+3.12\%/q$ vs $+2.45$; orthogonalized remainder $t = +0.40$; no nonlinearity, boosted–ridge $t = +0.27$): a third independent route rediscovering the same signal, and a lesson — evaluated on raw returns the ML predictions lose (inherited small-cap exposure): positioning factors must trade size-hedged, which NCB’s construction does by design.

Hyperparameter governance. Every parameter is justified by measurement or labeled a convention: the D+45 clock (delay distribution + ragged test), stress spec (published ablation surface), split thresholds (conservative; failure mode is a missed split, not a fabricated one), NCB surface (uniformly positive; selection valid), FSS threshold (monotone mechanism; price surface = regime, selection refused), LO tilt k (bootstrap IR distributions per k : in-sample ordering reverses out-of-sample, suggestive at IR-s.e. ≈ 0.4 ; an *adaptive* re-selector was simulated and picks $k=4$ throughout — $\approx$ holding the most aggressive tilt into the regime flip, -0.11 post-2021 vs $+0.31$ frozen: no evidence re-selection helps, weak evidence it hurts, so k ships as a tracking-error budget, default $k=1$). Across three applications of the bootstrap-selection protocol, its value was less picking parameters than *detecting when picking is illegitimate*.

Replications, stated as such. CHS breadth: replicated. Gompers–Metrick IO level and Chincarini dio/psd: dead after size, as the modern record says. Days-of-ADV: raw VW point replicated ($+1.46\%/mo$), dies residualized — the published effect is a size/liquidity exposure. Agarwal confidential holdings: does not replicate on 2013–25 ($t = -0.86$). Miori–Cucuringu imbalance reversal: oracle form replicates; bias-adjusted (point-in-time, churn-controlled) the premium is a fraction — who-knew-what-when accounting is most of the published edge. Coval–Stafford: mechanism replicates ($t \approx +10$); expected-flow *forecasting* does not transfer to 13F-implied flows. Risk monitors that survived: short-squeeze spikes revert ($-1.21\%/5d$ after, $t = -2.6$: the anti-panic rule is hold), and an aggregate deleveraging index conditions risk (corr $+0.14$ with future vol, zero with returns; gross throttle improves combo Sharpe $0.49 \rightarrow 0.55$) — state as conditioner, never signal.

Low confidence / noise vs signal. FSS’s price payoff rests on one 20-quarter regime — hedge, not all-weather alpha. IPCA’s edge and the sharper-NCB corner are candidates pending fresh sample. The LO tilt ordering and

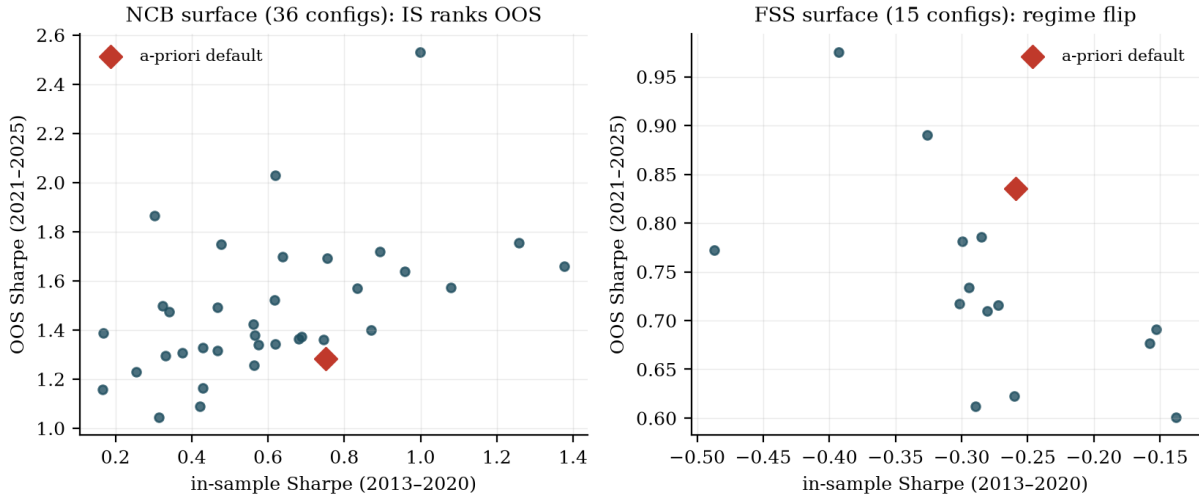


Figure 3: Hyperparameter governance in one picture: in-sample vs held-out Sharpe per configuration. Left, NCB: the whole surface is positive and in-sample ranks out-of-sample (+0.53) — selection is meaningful and the a-priori default (diamond) is conservative. Right, FSS: the grid moves as one block across a regime flip (−0.81) — selection is refused and the regime is reported.

Table 4: Signal catalogue, top and bottom (17 signals, headline version fixed ex-ante by the placebo rule; 49 quarters; full table in the report). The conviction family leads; states cluster at or below zero.

signal	version	spread/q	t	Sharpe	IC
NCB (new conviction)	resid	+0.0218	+3.54	1.08	+0.023
conviction top (level)	resid	+0.0223	+2.39	0.65	+0.024
breadth level	resid	+0.0193	+1.48	0.36	+0.027
entry rate	raw	+0.0134	+1.23	0.36	+0.004
Δ breadth (common)	raw	+0.0049	+1.07	0.27	+0.008
days-of-ADV (crowding)	resid	−0.0066	−0.41	−0.12	+0.012
holder concentration	resid	−0.0048	−1.09	−0.29	−0.012
Δ inst. ownership (2q)	raw	−0.0083	−1.89	−0.46	−0.009

the Q4 quarter-end anomaly are noise-until-replicated. Gross research spreads are systematically kinder than net engine results; every promoted claim above is the net one. The single strongest evidence that the survivors are real is procedural: the same funnel retracted its own marginal result when the data improved.

6. The three things we would build next

1. CRSP-tier prices with corporate-event unification. Closes the one measured open bias (survivorship truncations that cut against both legs), captures delisting returns the FSS mechanism partly trades, unifies trading items across ticker recycling and CUSIP reassignment via permno, lifts value coverage 85.4→~98% — and every headline is re-verified by the same campaign machinery that already caught one artifact. Highest inferential payoff per unit of work; everything downstream inherits it.

2. Probabilistic position nowcasting between filings, fed by higher-frequency flow/performance data. The 45-day staleness is the binding constraint on everything tactical; the oracle-vs-adjusted replication quantifies the prize (with true positions the imbalance premium is a multiple of the lagged version). The new data — monthly N-PORT portfolios plus Morningstar-tier reported TNA, subscriptions/redemptions, and returns — does two jobs. It feeds the nowcast alongside inputs already validated (drifted books, the flow instrument, real short interest supervising our synthetic proxy at corr 0.55, mechanical ETF rebalance rules); and it re-opens flow *prediction* honestly: forecasting failed on 13F-implied flow (leverage and measurement error destroy its autocorrelation) — the literature’s predictability lives in *reported* flows, with NMF-cluster hierarchical shrinkage as the waiting estimator. Deliverable: a probabilistic estimate of *current* holdings with honest uncertainty; gate, mechanism-first: beat the frozen-book baseline on holdings reconstruction OOS before any return claim.

3. Fundamentals and a risk model for residualization and false-discovery control. The placebo currently neutralizes size and liquidity — the exposures 13F counting is mechanically loaded on — but value/quality/profitability tilts of conviction buying remain unmodeled. Accounting data makes the theorem test characteristic-complete, turns crowding measurement into proper risk-model exposures, and supports *joint* false-discovery control across the catalogue instead of per-family Bonferroni: the difference between “NCB is not size” (shown) and “NCB is not any priced characteristic” (the next bar).